Cambridge
O Level

**Cambridge Assessment International Education**
Cambridge Ordinary Level

**CHEMISTRY** **5070/22**

Paper 2 Theory **October/November 2017**

MARK SCHEME

Maximum Mark: 75

**Published**

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| Question | Answer | Marks |
|---|---|---|
| A1(a)(i) | Iron / Fe (1) | **1** |
| A1(a)(ii) | Oxygen / $O_2$ (1) | **1** |
| A1(a)(iii) | Hydrogen / $H_2$ (1) | **1** |
| A1(a)(iv) | Aluminium / A*l* (1) | **1** |
| A1(a)(v) | Aluminium / A*l* (1) | **1** |
| A1(b) | $^{41}K$ electrons 19 (1) neutrons 22 (1)<br>$^{17}O^{2-}$ electrons 10 (1) neutrons 9 (1) | **4** |

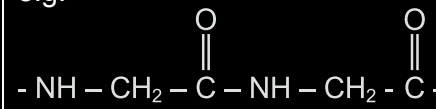
| Question | Answer | Marks |
|---|---|---|
| A2(a) | Arrangement: ordered / lattice / regular / layers / uniformly arranged / repeated pattern (1)<br><br>Type of force: electrostatic (1) | **2** |
| A2(b) | Crystals: ions cannot move (1)<br><br>Aqueous: ions can move (1) | **2** |
| A2(c) | Magnesium ion 2,8 (1)<br>Chloride ion 2,8,8 (1) | **2** |
| A2(d)(i) | Hydrogen is lower in the reactivity series (than sodium) / sodium more reactive (than hydrogen) / hydrogen ions are reduced more easily (than sodium ions) (1) | **1** |
| A2(d)(ii) | $2H^+ + 2e^- \rightarrow H_2$ (1) | **1** |
| A2(d)(iii) | Litmus paper / named indicator paper (1)<br><br>Bleaches / loses its colour (1) | **2** |

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| Question | Answer | Marks |
|---|---|---|
| A2(e)(i) | All four of: $H^+$, $OH^-$, $Cl^-$, $Na^+$ (1) | **1** |
| A2(e)(ii) | Hydrogen ions removed (1)<br><br>Idea of $OH^-$ remaining (causing alkalinity) (1) | **2** |

| Question | Answer | Marks |
|---|---|---|
| A3(a) | 1 mark each for any two of:<br>• Shiny / lustrous<br>• Ductile<br>• Malleable | **2** |
| A3(b) | Iron<magnesium<cerium<sodium (1) | **1** |
| A3(c)(i) | Mol Fe = $\frac{39.2}{56}$ **OR** 0.7(00) (1)<br><br>Mol $Fe_3O_4$ = $\frac{0.7(00)}{3}$ **OR** 0.233 (1)<br><br>Mass = $0.233 \times 232 = 54.1$ (1) | **3** |
| A3(c)(ii) | Moles $H_2 = 4 \times 0.233$ **OR** 0.933 (1)<br><br>Volume = $0.933 \times 24 = 22.4\ dm^3$ (1) | **2** |
| A3(d) | CO is poisonous / toxic (1) | **1** |

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| Question | Answer | Marks |
|---|---|---|
| A4(a)(i) | $C_{10}H_{18}O$ (1) | **1** |
| A4(a)(ii) | Alcohol (1) | **1** |
| A4(b) | Bromine / bromine water (1)<br><br>Turns colourless (1) | **2** |
| A4(c) | 1 mark each for any three of:<br><br>• Pigment(s) / dyes / coloured compounds on paper and paper (dipping) in solvent<br><br>• Spot of dye above solvent level<br><br>• Measure distance moved by dye and solvent (front)<br><br>• Calculate $R_f$ value<br><br>• Compare with known $R_f$ value(s) | **3** |
| A4(d) | Two correct repeat units with amide link (2 marks)<br>e.g.<br>  <br><br>If 2 marks not awarded, 1 mark for two amide links drawn correctly | **2** |

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| Question | Answer | Marks |
|---|---|---|
| A5(a) | $2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + CO_2 + H_2O$<br>Correct formulae (1)<br>Correct balance (1) | **2** |
| A5(b) | Bond breaking is endothermic and bond making is exothermic (1)<br>Less energy released (in bond making) than absorbed (in bond breaking) (1) | **2** |
| A5(c) | Solvents / flavourings / perfumes (1) | **1** |

| Question | Answer | Marks |
|---|---|---|
| B6(a)(i) | No effect (1)<br>Equal number of moles (of gases) on each side of the equation / each volumes (of gases) on each side of the equation (1) | **2** |
| B6(a)(ii) | Equilibrium moves to the left (1)<br>The (forward) reaction is endothermic / backward reaction exothermic / goes in the direction of the exothermic reaction (1) | **2** |
| B6(b)(i) | Substance which completely dissociates to form $H^+$ ions / substance which completely ionises to form $H^+$ ions | **1** |
| B6(b)(ii) | $HI \rightarrow H^+ + I^-$ (1) | **1** |
| B6(c) | Addition (1) | **1** |
| B6(d)(i) | Increases (as the number of C atoms increases) (1) | **1** |
| B6(d)(ii) | Liquid because –7 °C / it is above its melting point and below its boiling point / liquid because –7 °C / it is between the melting point and boiling point (1) | **1** |
| B6(d)(iii) | There is no clear trend / the values go up and down / the values go down and up (1) | **1** |

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| Question | Answer | Marks |
|---|---|---|
| B7(a)(i) | Magnesium has strong bonding between positive ions / cations and electrons / magnesium is a giant structure (1)<br>Sulfur is a simple molecule / weak forces between sulfur molecules (1) | **2** |
| B7(a)(ii) | Magnesium has electrons which move (from place to place) (1)<br>Sulfur does not have delocalised electrons / no mobile electrons / electrons don't move (1) | **2** |
| B7(b) | Giant structure / many covalent bonds (1)<br>Need high temperature / lot of energy to break the bonds (1) | **2** |
| B7(c)(i) | Mass of sulfur = 19.2 g (1)<br>mol S = $\frac{19.2}{32}$ mol $Cl$ = $\frac{21.3}{35.5}$ **OR** ratio = 0.6 to 0.6 (1)<br>S$Cl$ (1) | **3** |
| B7(c)(ii) | $S_2Cl_2$ (1) | **1** |

| Question | Answer | Marks |
|---|---|---|
| B8(a) | $\frac{2 \times 39}{174} \times 100$ = 44.8% / 45% (2 marks)<br>If 2 marks not scored correct $M_r$ = 174 (1) | **2** |
| B8(b) | (Acidified) barium chloride / barium nitrate (1)<br>White precipitate (1) | **2** |
| B8(c) | Nitrates soluble (in water) / nitrates dissolve (easily) (1) | **1** |

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| Question | Answer | Marks |
|---|---|---|
| B8(d) | One mark each for any two of:<br>• (Nitrate causing) excessive growth of algae<br>• Bacterial growth (on dead algae)<br>• Idea of making water deoxygenated so animals / organisms can't live | **2** |
| B8(e) | Mol KOH = $0.200 \times \frac{12.5}{1000}$ **OR** $2.5 \times 10^{-3}$ (1)<br>Mol phosphoric acid = $\frac{2.5 \times 10^{-3}}{3}$ **OR** $8.33 \times 10^{-4}$ (1)<br>Concentration of phosphoric acid = 0.0333 (mol / $dm^3$) (1)<br>($8.33 \times 10^{-4} \times 1000 / 25$) | **3** |

| Question | Answer | Marks |
|---|---|---|
| B9(a)(i) | Rate doubles as concentration doubles (or reverse argument) (1) | **1** |
| B9(a)(ii) | Particles closer together / more particles in a given volume (or reverse argument) (1)<br>Higher collision frequency / more collisions per second (or reverse argument) (1) | **2** |
| B9(a)(iii) | Increasing temperature increases rate (no marks)<br>Particles move faster / particles have greater kinetic energy (1)<br>Number of particles with energy greater than the activation energy is increased / more successful collisions / more effective collisions (1) | **2** |

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| Question | Answer | Marks |
|---|---|---|
| B9(b)(i) | Burning fossil fuels / burning named fossil fuel / volcanoes (1) | **1** |
| B9(b)(ii) | It is reformed / it is not used up / it is unchanged at the end of the reaction (1) | **1** |
| B9(c)(i) | Energy humps of both the uncatalysed and catalysed reaction shown and labelled with catalysed reaction below the uncatalysed and hump drawn correctly form reactants line to product line. (2 marks)<br><br>If 2 marks not scored allow 1 mark for one or two energy humps drawn correctly from reactants line to products line (1) | **2** |
| B9(c)(ii) | Exothermic because energy of reactant greater than energy of products (or reverse argument) (1) | **1** |

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